

Sigma-Unital Coronas Fail the Ball-Covering Property

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when K has a countable π -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

Let A be a nonunital C^* -algebra. Write $\mathcal{M}(A)$ for its multiplier algebra and

$$Q : \mathcal{M}(A) \longrightarrow \mathcal{M}(A)/A$$

for the quotient map.

Theorem 1. *If A is nonunital and σ -unital, then the corona*

$$\mathcal{M}(A)/A$$

fails the ball-covering property.

Corollary 1. *Let D be a nonzero σ -unital C^* -algebra and let H be an infinite-dimensional separable Hilbert space. Then*

$$\mathcal{M}(\mathcal{K}(H) \otimes_{\min} D) / (\mathcal{K}(H) \otimes_{\min} D)$$

fails BCP.

Proof. The algebra $\mathcal{K}(H) \otimes_{\min} D$ is nonunital and σ -unital, so Theorem 1 applies. □

Idea of the Proof

The proof replaces finite coordinate projections by annuli in a sequential approximate unit. Choose a positive contractive approximate unit (e_n) with $e_{n+1}e_n = e_n$. For $T \in \mathcal{M}(A)$, the quotient norm $\|Q(T)\|$ is visible on right tails $T(1 - e_n)$, and then on finite annuli $T(e_m - e_n)$. Given countably many proposed centers, recursively choose disjoint inner annuli where the assigned center nearly realizes its quotient norm. Around each inner annulus put a slightly larger annulus and choose a scalar phase that is Hahn–Banach far from the local piece of the center. The strict sum of the phased outer annuli is a multiplier of norm at most one. Its corona class has norm one and avoids every proposed ball $B(\xi, \|\xi\|)$.

Preliminaries

We use the standard equivalent form of BCP: if a normed space Y has BCP, then S_Y is covered by countably many open balls $B(y_j, \|y_j\|)$.

Since A is σ -unital, fix a positive contractive approximate unit (e_n) satisfying

$$e_{n+1}e_n = e_n, \quad n \geq 1.$$

Passing to later terms gives $e_me_n = e_n$ whenever $m \geq n$. In particular the e_n 's commute and $e_m - e_n$ is a positive contraction whenever $m \geq n$.

Lemma 1 (quotient norm on approximate-unit tails). *For every $T \in \mathcal{M}(A)$,*

$$\|Q(T)\| = \lim_{n \rightarrow \infty} \|T(1 - e_n)\|.$$

Proof. Since $Te_n \in A$, we have $\|Q(T)\| \leq \|T(1 - e_n)\|$ for every n . Conversely, if $C \in A$, then $C(1 - e_n) \rightarrow 0$ in norm, and hence

$$\limsup_n \|T(1 - e_n)\| \leq \limsup_n \|(T - C)(1 - e_n)\| + \limsup_n \|C(1 - e_n)\| \leq \|T - C\|.$$

Taking the infimum over $C \in A$ gives the reverse inequality. □

Lemma 2 (finite annulus recovery). *Let $T \in \mathcal{M}(A)$, put $r = \|Q(T)\|$, and let $N \in \mathbb{N}$ and $\varepsilon > 0$. Then there are integers*

$$N < n < m$$

such that

$$\|T(e_m - e_n)\| > r - \varepsilon.$$

Proof. By Lemma 1, choose $n > N$ such that

$$\|T(1 - e_n)\| > r - \varepsilon/2.$$

For every $B \in \mathcal{M}(A)$,

$$\|B\| = \sup_m \|Be_m\|.$$

Indeed, the inequality “ $\geq$ ” is immediate, while for $a \in A$ with $\|a\| \leq 1$,

$$\|Ba\| = \lim_m \|Be_ma\| \leq \sup_m \|Be_m\|,$$

and A is an essential ideal in $\mathcal{M}(A)$. Applying this to $B = T(1 - e_n)$, choose $m > n$ so that

$$\|T(1 - e_n)e_m\| > r - \varepsilon.$$

Since $e_n e_m = e_n$, this is exactly

$$\|T(e_m - e_n)\| > r - \varepsilon.$$

□

Lemma 3 (orthogonal strict sums). *Let*

$$a_k < n_k < m_k < b_k < a_{k+1}, \quad k \geq 1,$$

and define

$$g_k = e_{m_k} - e_{n_k}, \quad h_k = e_{b_k} - e_{a_k}.$$

Then the h_k 's are pairwise orthogonal positive contractions,

$$h_k g_k = g_k, \quad h_j g_k = 0 \quad (j \neq k),$$

and for every bounded scalar sequence (λ_k) , the series

$$z = \sum_{k=1}^{\infty} \lambda_k h_k$$

converges strictly to an element of $\mathcal{M}(A)$ with

$$\|z\| \leq \sup_k |\lambda_k|, \quad z g_k = \lambda_k g_k.$$

Proof. All identities follow from $e_m e_n = e_{\min(m,n)}$ for m, n in the chosen subsequence. For example, if $a_k < n_k < m_k < b_k$, then

$$(e_{b_k} - e_{a_k})(e_{m_k} - e_{n_k}) = e_{m_k} - e_{n_k} = g_k.$$

If the intervals are disjoint, the same computation gives $h_j h_k = 0$ and $h_j g_k = 0$ for $j \neq k$.

For strict convergence, let $s_N = \sum_{k \leq N} \lambda_k h_k$. The partial sums are uniformly bounded by $\sup_k |\lambda_k|$, because the h_k 's are orthogonal positive contractions. If $a \in A$, then the tail $\sum_{k > N} \lambda_k h_k$ is annihilated by $e_{a_{N+1}}$, so

$$\left\| \left(\sum_{k > N} \lambda_k h_k \right) a \right\| = \left\| \left(\sum_{k > N} \lambda_k h_k \right) (a - e_{a_{N+1}} a) \right\| \leq \sup_k |\lambda_k| \|a - e_{a_{N+1}} a\| \rightarrow 0.$$

The same argument on the other side gives

$$\left\| a \left(\sum_{k > N} \lambda_k h_k \right) \right\| \rightarrow 0.$$

Thus the series is strictly convergent in $\mathcal{M}(A)$, with the stated norm bound and block action. □

Proof of the Theorem

Proof of Theorem 1. Let

$$\xi_i = Q(T_i), \quad i \geq 1,$$

be an arbitrary countable family of centers in $\mathcal{M}(A)/A$, with $T_i \in \mathcal{M}(A)$. Put $r_i = \|\xi_i\|$. Add one auxiliary multiplier

$$T_0 = 1_{\mathcal{M}(A)}, \quad r_0 = 1.$$

Choose a map $\pi : \mathbb{N} \rightarrow \{0, 1, 2, \dots\}$ such that every value occurs infinitely often.

We recursively choose integers

$$a_k < n_k < m_k < b_k < a_{k+1}$$

and phases $\lambda_k \in \mathbb{T}$. Suppose the previous choices have been made, and put $i = \pi(k)$. Choose $a_k > b_{k-1}$, with the convention $b_0 = 0$. By Lemma 2, choose

$$a_k < n_k < m_k$$

so that, with $g_k = e_{m_k} - e_{n_k}$,

$$\|T_i g_k\| > r_i - \frac{1}{k}.$$

Choose $b_k > m_k$, and later choose $a_{k+1} > b_k$.

Now choose $\lambda_k \in \mathbb{T}$ so that

$$\|\lambda_k g_k - T_i g_k\| \geq \|T_i g_k\|.$$

This is just the Hahn–Banach sign trick: if $T_i g_k \neq 0$, take a norm-one functional φ with $\varphi(T_i g_k) = \|T_i g_k\|$, and choose $\lambda_k \in \mathbb{T}$ with $\operatorname{Re}(\lambda_k \varphi(g_k)) \leq 0$. If $T_i g_k = 0$, any λ_k works.

Let

$$h_k = e_{b_k} - e_{a_k}.$$

By Lemma 3, the series

$$z = \sum_{k=1}^{\infty} \lambda_k h_k$$

converges strictly to a multiplier with $\|z\| \leq 1$, and

$$z g_k = \lambda_k g_k.$$

We first show that $\|Q(z)\| = 1$. The upper bound follows from $\|z\| \leq 1$. Let $C \in A$. Take $k \rightarrow \infty$ along the infinite set $\{k : \pi(k) = 0\}$. For these k 's,

$$\|g_k\| = \|T_0 g_k\| > 1 - \frac{1}{k}.$$

Moreover $n_k \rightarrow \infty$, so

$$\|C g_k\| = \|C(1 - e_{n_k})e_{m_k}\| \leq \|C(1 - e_{n_k})\| \rightarrow 0.$$

Therefore

$$\|z - C\| \geq \|(z - C)g_k\| \geq \|z g_k\| - \|C g_k\| = \|g_k\| - \|C g_k\|,$$

and letting $k \rightarrow \infty$ along $\pi(k) = 0$ gives $\|z - C\| \geq 1$. Taking the infimum over $C \in A$, we get $\|Q(z)\| \geq 1$.

Now fix $i \geq 1$ and $C \in A$. Let $k \rightarrow \infty$ along the infinite set $\{k : \pi(k) = i\}$. Since $n_k \rightarrow \infty$, $\|Cg_k\| \rightarrow 0$. Hence

$$\|z - T_i - C\| \geq \|(z - T_i - C)g_k\| \geq \|\lambda_k g_k - T_i g_k\| - \|Cg_k\| \geq \|T_i g_k\| - \|Cg_k\| > r_i - \frac{1}{k} - \|Cg_k\|.$$

Letting $k \rightarrow \infty$ along this subsequence gives

$$\|z - T_i - C\| \geq r_i.$$

Taking the infimum over $C \in A$, we obtain

$$\|Q(z) - \xi_i\| \geq r_i = \|\xi_i\|.$$

Thus $Q(z)$ is a unit vector in $\mathcal{M}(A)/A$ outside every open ball

$$B(\xi_i, \|\xi_i\|).$$

Since the countable family of centers was arbitrary, $\mathcal{M}(A)/A$ fails BCP. □

Scope and Relation to Earlier Packets

The previous stabilized-corona packet proved the result for

$$(\mathcal{B}(H) \otimes D)/(\mathcal{K}(H) \otimes D)$$

with unital coefficient algebra D , using finite-rank coordinate projections and partial isometries. The present theorem is both simpler and broader for multiplier coronas: it applies to every nonunital σ -unital C^* -algebra A . In particular it covers

$$A = \mathcal{K}(H) \otimes_{\min} D$$

for every nonzero σ -unital coefficient algebra D , unital or not.

The theorem is still a partial answer to the source question. It does not characterize arbitrary C^* -algebras with BCP, and it does not address coronas of non- σ -unital ideals.

Verification and Search Bounds

The proof is self-contained modulo the standard fact that a σ -unital C^* -algebra has a positive contractive approximate unit (e_n) satisfying $e_{n+1}e_n = e_n$. The run indexes were searched for arXiv:2311.14257 together with BCP, UBCP, corona, multiplier algebra, σ -unital, approximate unit, Calkin, and stable multiplier keywords. Existing run packets covered product quotients, reduced products, continuous fields, unital stabilized coronas, and a Hilbert-module block criterion, but no packet contained this approximate-unit annulus theorem.

External literature checks during this push found arXiv:2412.07137, which proves that several Calkin algebras fail BCP, and arXiv:2111.04921, which gives commutative/vector-valued BCP stability results and warns that BCP is not hereditary even for 1-complemented subspaces. Those papers are consistent with the present theorem but do not state the sigma-unital multiplier-corona annulus result used here.

Human Review Focus

Review Lemma 2 and Lemma 3. These are the two structural points: quotient norms must be recoverable on finite approximate-unit annuli, and the phased outer annuli must define a strict multiplier acting as the chosen phase on the inner annulus. Once those are checked, the final diagonal escape is formal.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Minzeng Liu, Rui Liu, Jimeng Lu, and Bentuo Zheng, *Ball covering property from commutative function spaces to non-commutative spaces of operators*, arXiv:2111.04921, 2021.
- [3] Qiyao Bao, Rui Liu, and Jie Shen, *The ball-covering property of non-commutative spaces of operators on Banach spaces*, arXiv:2412.07137, 2024.